

Chenyang LIU

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EDUCATION

Duke University

Durham, USA

PhD in Computer Science, advised by Prof. Matthew Lentz and Prof. Kartik Nayak Aug. 2024 – Present

Shanghai Jiao Tong University

Shanghai, China

Bachelor of Engineering in Computer Science, GPA: 90.33/100, Rank: 3/86 Sep. 2020 – June 2024

RESEARCH INTERESTS

My research lies at the intersection of distributed systems, blockchain, and security. I design and implement Byzantine Fault-Tolerant (BFT) consensus protocols that are both secure and efficient; my current work explores signature-free BFT protocols that remain secure in the post-quantum era. I also build confidential computing systems on Trusted Execution Environments (TEEs), from decentralizing Ethereum's proposer-builder separation (PBS) to mitigating side-channel attacks.

RESEARCH EXPERIENCE

Graduate Research Assistant

Aug. 2024 – Present

Duke University

Durham, USA

DPaaS: Decentralized Proposer-as-a-Service for Ethereum PBS

- **Protocol Design:** Formalized the security properties of Ethereum PBS and designed a side-car protocol that outsources the combined roles of relay and proposer to a group of TEEs.
- **New Primitive Crab-BB:** Designed a generalized Byzantine Broadcast protocol with censorship resistance and backed availability for arbitrary applications, including Ethereum PBS.
- **Implementation and Evaluation:** Built a Rust prototype deployed on AWS, Azure, and GCP confidential VMs (AMD SEV-SNP, Intel TDX), showing 3% higher MEV earnings than relays.

Signature-Free BFT Consensus for the Post-Quantum Era

- **Protocol Design:** Designed PBFT-style information-theoretic BFT protocols for $n = 5f + 1$ with two-round good-case commit (matching signature-based protocols), $O(n^2)$ messages, and optimistic responsiveness.

Undergraduate Research Assistant

Feb. 2023 – July 2024

Shanghai Jiao Tong University, advised by Prof. Guoxing Chen

Shanghai, China

Closing Cache Side Channels via Minimal Cache Fingerprints

- **Cache-Set Flattening:** Designed the first flattening-based defense against cache side channels on Intel SGX, confining all sensitive code and data to a single cache set so that access traces reveal no secrets.
- **Compiler and Driver Support:** Built LLVM backend passes that split enclave code/data into 64-byte blocks and instrument memory accesses, plus an SGX driver for cache-color-aware page allocation.
- **Verifiable OS Contract:** Proposed *contrived self-eviction*, a timing-based test that lets the enclave verify the untrusted OS's page allocation on both inclusive and non-inclusive caches.

- **Evaluation:** Defeated Prime+Probe attacks on OpenSSL AES and GnuPG ElGamal with $3.25\times$ overhead on SGX-nbench; optimized contract verification to cut wolfSSL overhead from $12\times$ to $1.24\times$.

Formally Verified Privileged Enclaves

- **Case Studies and Evaluation:** Implemented three case studies and evaluated the performance of the proposed design on the Penglai RISC-V TEE platform.

PUBLICATIONS

- Quintus: Two-round Good-case Information Theoretic BFT for $n = 5f + 1$** In Submission
Chenyang Liu, Dahlia Malkhi, Kartik Nayak, Nibesh Shrestha
- DPaaS: Improving Decentralization and Trustworthiness in Ethereum PBS** In Submission
Chenyang Liu, Arnav Jindal, Ittai Abraham, Matthew Lentz, Kartik Nayak
- MINCE: Closing Cache Side Channels via Minimal Cache Fingerprints** Undergraduate Thesis
Chenyang Liu, Guoxing Chen, Jiaping Gui, Haojin Zhu
- A Formal Approach to Multi-Layered Privileges for Enclaves** NDSS 2025
Ganxiang Yang, Chenyang Liu, Zhen Huang, Guoxing Chen, Hongfei Fu, Yuanyuan Zhang, Haojin Zhu

TEACHING & SERVICE

- Teaching Assistant:** Intro. to Operating Systems (Spring 2025); Operating Systems (Fall 2025)
- Artifact Evaluation Committee:** ACM CCS 2027

SELECTED HONORS & AWARDS

- Flashbots MEV Fellowship Grant:** As student researcher 2025
- Outstanding Graduate:** Shanghai Jiao Tong University 2024
- Zhiyuan Honor Awards:** Top 5% of students in Zhiyuan Honor Program 2021 – 2023
- First Class Network Security Awards:** For contributions to security research 2023
- Academic Progress Awards:** For outstanding academic progress (3/88) 2022

SKILLS

- Programming:** Rust, C/C++, Python, x86/RISC-V Assembly, SQL, JavaScript, MATLAB
- TEEs:** Intel SGX, AMD SEV-SNP, Intel TDX, Penglai (RISC-V)
- Frameworks & Tools:** Git, LaTeX, Docker, LLVM, QEMU, Django, Google/AWS/Azure Cloud
- Languages:** Chinese (Native), English